Suppose we have k populations with means $\mu_1, \mu_2, \cdots, \mu_k$ and common variance σ^2.

Then let $H_0 := \mu_1 = \mu_2 = \cdots = \mu_k$ and let H_a be that at least one of the above inequality does not hold.

To test hypothesis of above form we use ANOVA (Analysis of Variance).

Suppose from i_{th} population, we collect a random sample of size n_i.

Let $n = n_1 + n_2 + \cdots + n_k$

Also, we denote the random sample from from first population as,

$$Y_{11}, Y_{12}, Y_{13}, \cdots, Y_{1n_1}$$

and similarly the random sample from i_{th} population as,

$$Y_{i1}, Y_{i2}, Y_{i3}, \cdots, Y_{in_i}.$$

Now let,

$$\overline{Y} = \frac{1}{n}\left(\sum_{i=1}^{k}\sum_{j=1}^{n_i} Y_{ij}\right)$$

Alternatively,

Let

$$\overline{Y_i} = \frac{1}{n_i}\left(\sum_{j=1}^{n_i} Y_{ij}\right)$$

Then

$$\overline{Y} = \frac{\overline{Y_1} + \overline{Y_2} + \cdots + \overline{Y_k}}{k}$$

Now we define, Total Sum of Squares, as,

$$\text{Total SS} = \sum_{i=1}^{k}\sum_{j=1}^{n_i}\left(Y_{ij} - \overline{Y}\right)^2$$

, Sum of Squares of Treatments, as

$$SST = \sum_{i=1}^{k} n_i\left(\overline{Y_i} - \overline{Y}\right)^2$$

, and Sum of Squares of Error, as,

$$SSE = \text{Total } SS - SST$$

or alternatively,

$$SSE = \sum_{i=1}^{k} \sum_{j=1}^{n_i} (Y_{ij} - \overline{Y_i})^2$$

It follows that, defining

$$MST = \frac{SST}{k-1}$$

follows χ^2 distribution with $(k-1)$ df

and

$$MSE = \frac{SSE}{(n-k)}$$

follows χ^2 distribution with $(n-k)$ df.

Then defining,

$$F = \frac{MST}{MSE}$$

we have that it follows F distrib-ution with $(k-1)$ numerator df and $(n-k)$ denominator df.

Finally, to test, H_0 vs H_a as defined earlier, with significance level α, using the test statistic

$$F = \frac{MST}{MSE}$$

we define rejection region as,

$$F > F_\alpha .$$

Eg:

Four groups of students were subjected to different teaching techniques and tested at the end of a specified period of time. As a result of dropouts from the experimental groups, the number of students varied from group to group. Do the data shown below present sufficient evidence to indicate a difference in mean achievement for the four teaching techniques? Test at significance level $\alpha = 0.05$.

1	2	3	4
65	75	59	94
87	69	78	89
73	83	67	80
79	81	62	88
81	72	83	
69	79	76	
	90		

From the data in the above table we compute,

$$\bar{y}_1 = 75.67$$

$$\bar{y}_2 = 78.43$$

$$\bar{y}_3 = 70.83$$

$$\bar{y}_4 = 87.75$$

$$\bar{y} = 77.35$$

Then,

Total SS = 1909.217

SST = 712.5864

SSE = 1196.631

and

$$MST = 237.5288$$

$$MSE = 62.98058$$

Finally,

$$f = \frac{237.5288}{62.98058} = 3.771461$$

and,

$$F_{0.05} = qf(0.95, 3, 19) = 3.12$$

Since, $3.77 > 3.12$

therefore we reject the null hypothesis.